Table of Contents

iMotions Online

Webcam Eye Tracking 2

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

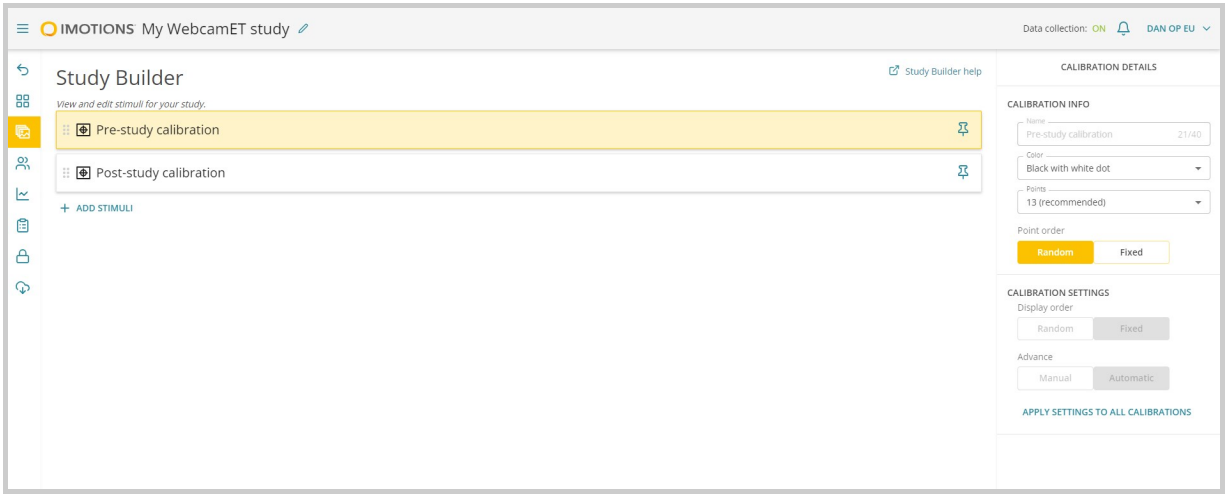
Webcam Eye Tracking

This article describes study setup, data collection, and visualization and analysis of Webcam-based eye tracking using iMotions Online and the iMotions WebET 3.0 algorithm (link to white paper).

Setting up your study

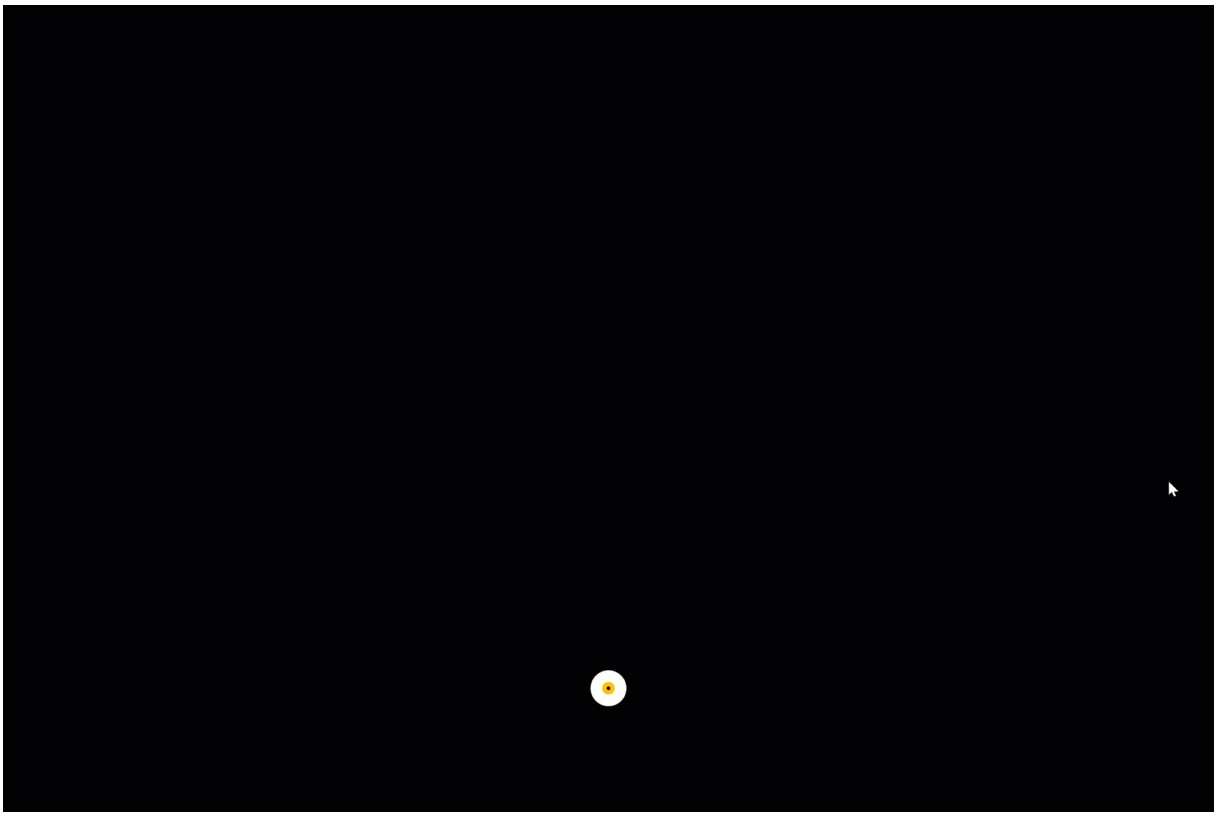
Webcam-based eye tracking data is automatically collected for all stimuli in iMotions Online studies.

Eye tracker calibration is automatically included, as the "Pre-study calibration" and "Post-study calibration" stimuli. Each calibration stimulus will display calibration points, according to the setting in Calibration Details.

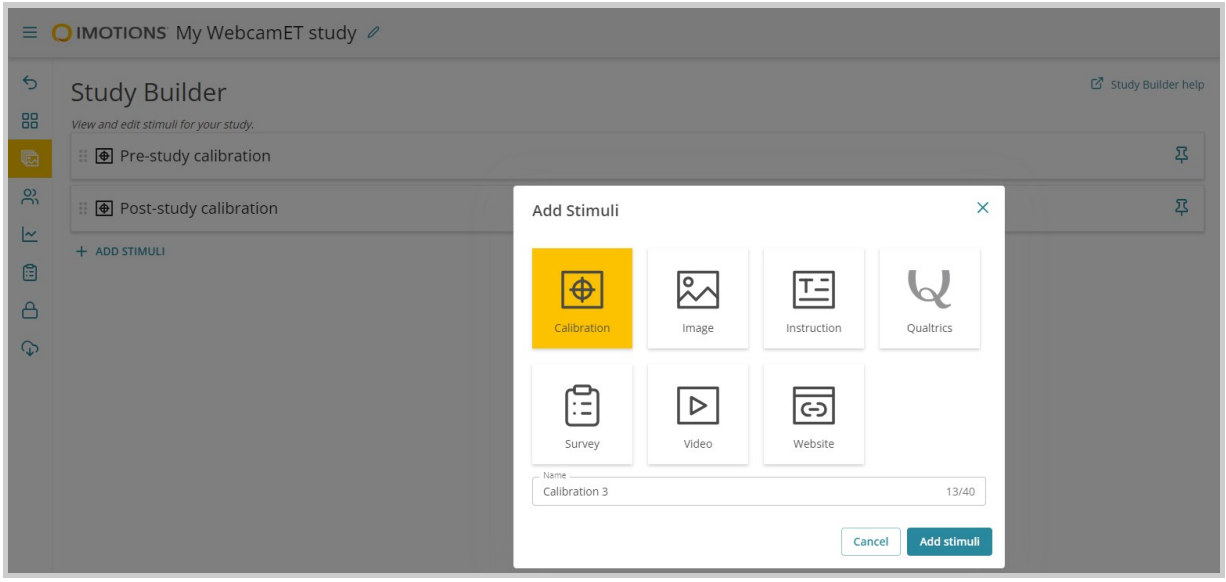


Example of calibration stimulus:

Confidential © 2023 Redistribution is not permitted without written permission from iMotions



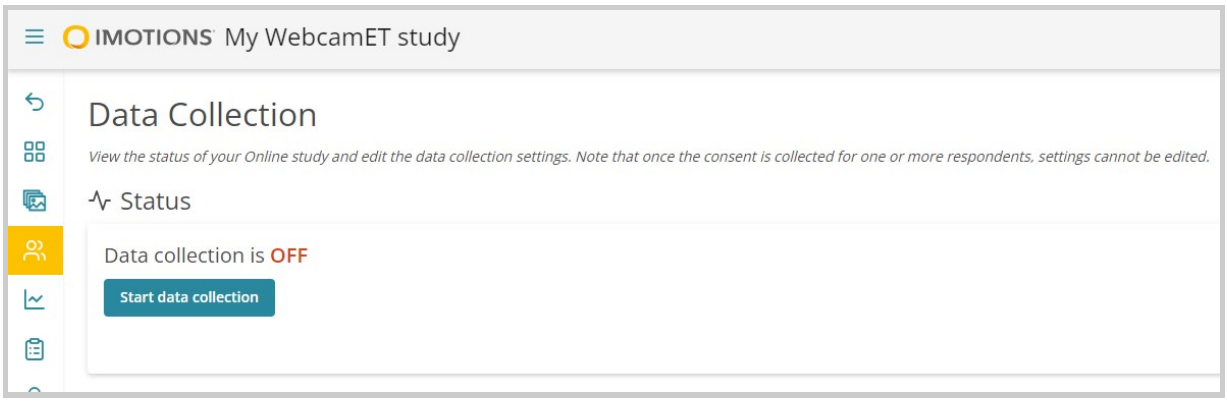
Additional Calibration stimuli can be added to studies, depending on the duration of the study or any other demands for additional calibration data (e.g. studies over 10 minutes, respondent populations prone to movement, before/after critical stimuli).



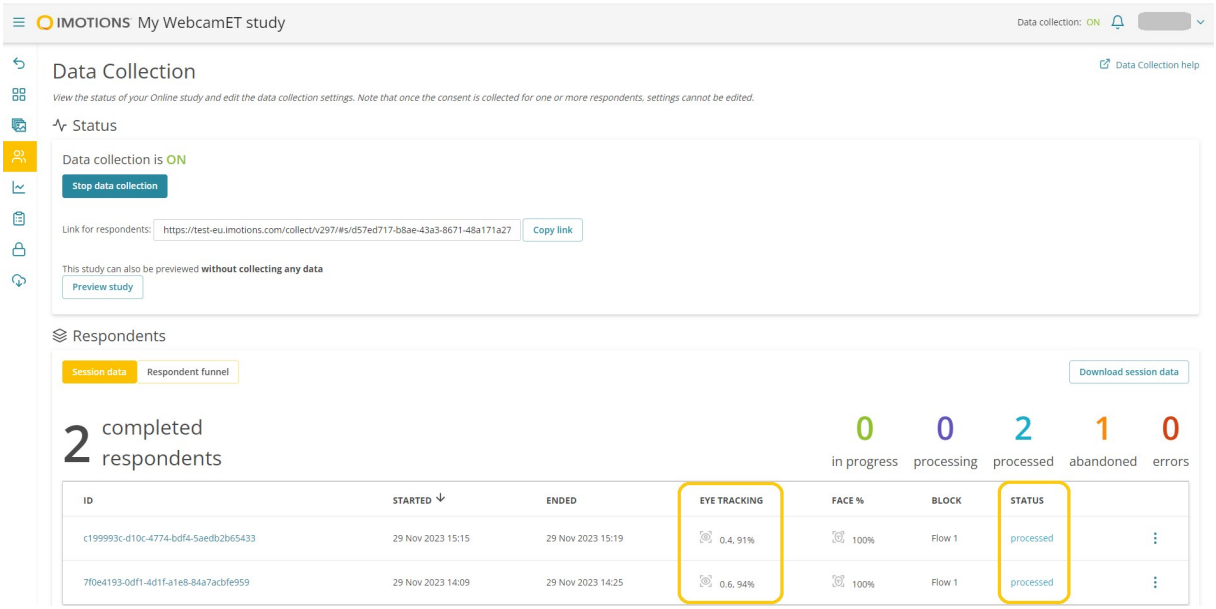
Data collection

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

Once the study is ready, on the Data Collection tab, click "Start data collection" to switch Data collection to ON and create the "Link for respondents".



Opening the link will initiate study participation, including the eye-tracking data collection. Once the study is completed, the respondent data will be uploaded and processed. Once the data are processed, the respondent Status will change from "processing" to "processed" and data quality metrics will appear under "Eye Tracking", which indicates eye tracking accuracy (in degrees) and quality (% of recorded data/expected data).



Visualization and Analysis

Overview of Analysis with iMotions Online

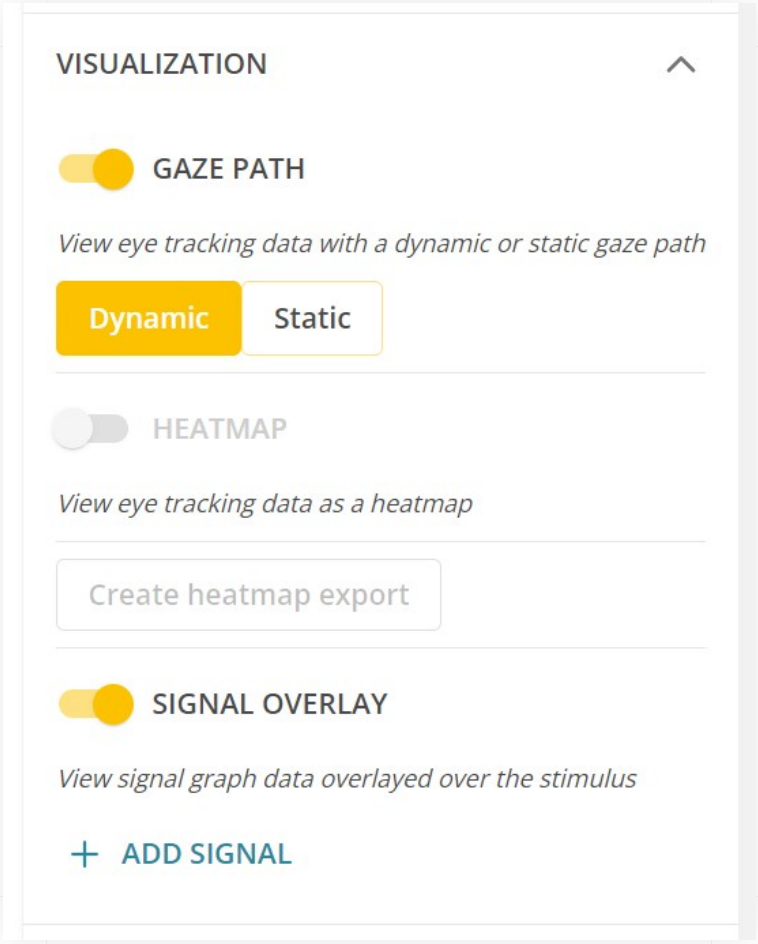
The iMotions Online Analysis page has built-in tools for visualizing and analyzing eye tracking data:

- Gaze Paths for Individual respondents (with or without signal overlay)

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

- Heat Maps for Aggregate groups of respondents
- Area of Interest (AOI) analysis

Eye Tracking Visualization menu:



Example Gaze Path visualization:

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

Image 1 & 0064cd34-ed09-4d6a-9a5b-d07303db8948



Example Gaze path visualization with signal overlay:

Image 1 & 0064cd34-ed09-4d6a-9a5b-d07303db8948



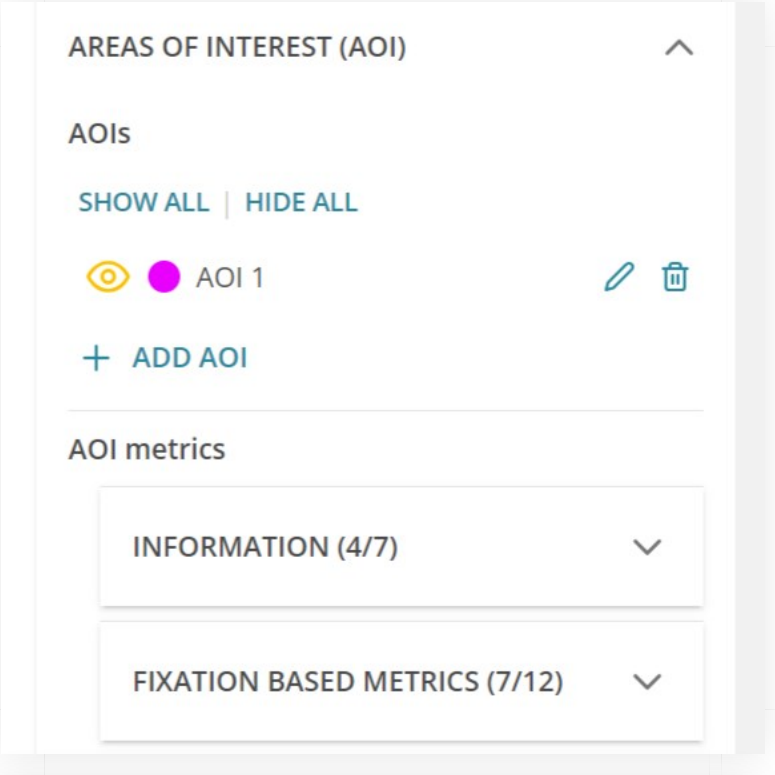
Example Heat Map:

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

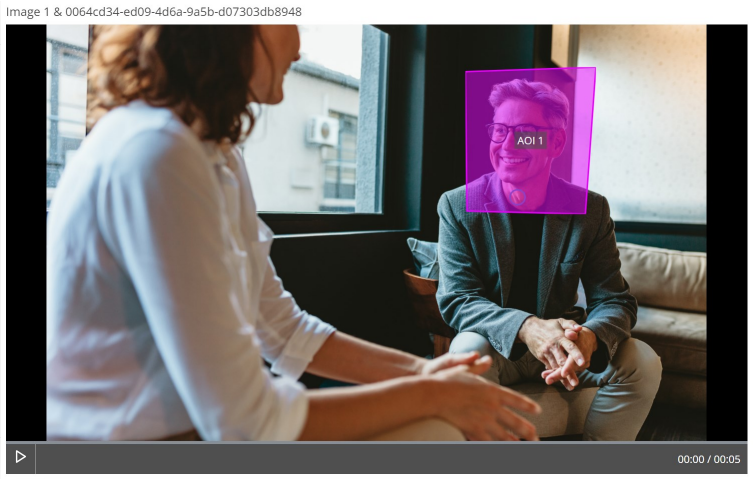


The Area of Interest (AOI) interface allows the definition of AOIs and the calculation of AOI metrics. Once the AOI metrics are calculated (or updated, when AOIs are edited or added), the "Download AOI metrics" button/interface can then be used to obtain a CSV file containing these results.

Example AOI Definition:

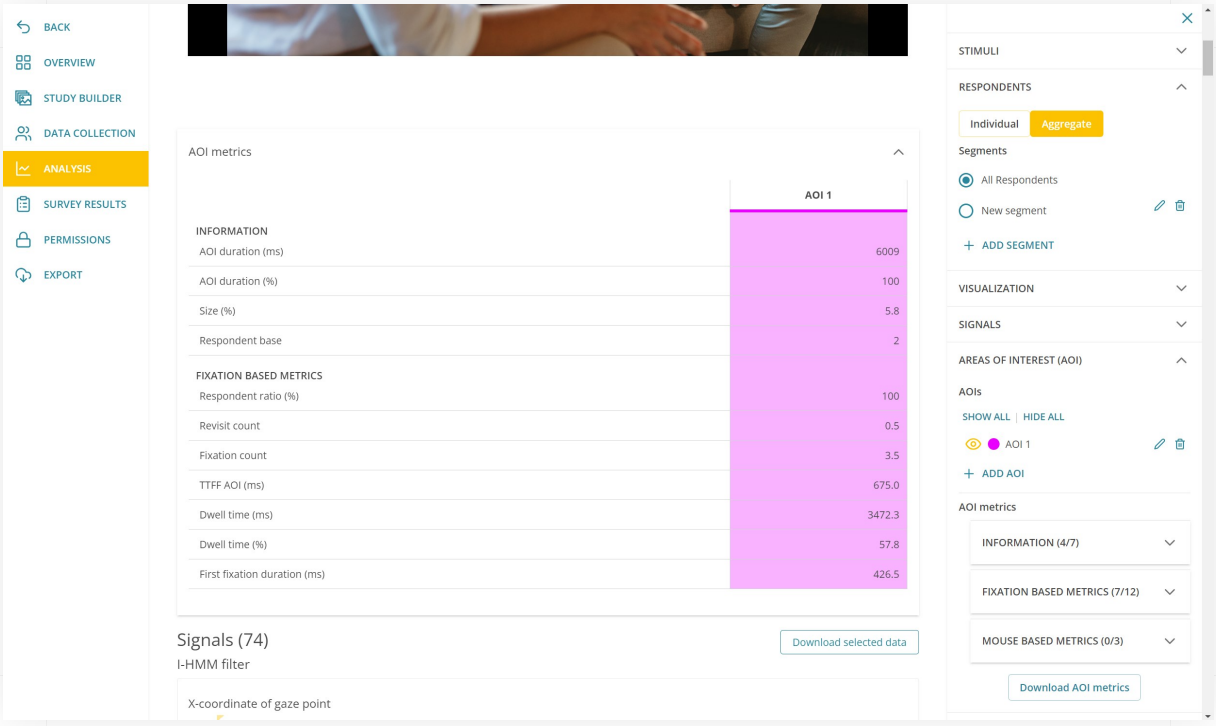


Confidential © 2023 Redistribution is not permitted without written permission from iMotions



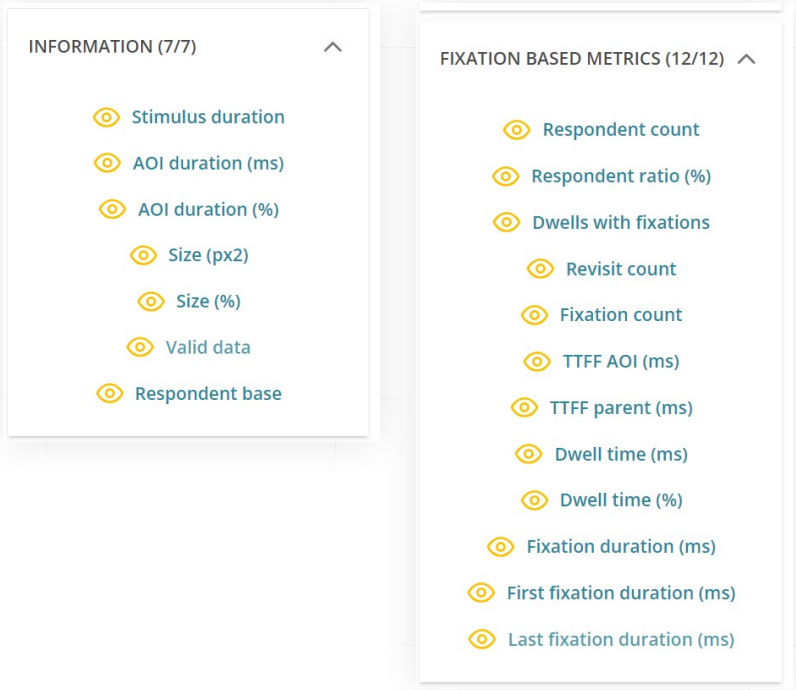


Example AOI Metrics:



Confidential © 2023 Redistribution is not permitted without written permission from iMotions

List of AOI Metrics:



Powered by Document360

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

Confidential © 2023 Redistribution is not permitted without written permission from iMotions